

FAPS: Failure-Aware Policy Shielding for Vision-Language-Action Models

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Motivation

VLA models generalize across manipulation tasks but remain brittle under distribution shift. Sensor noise, viewpoint drift, and actuation jitter corrupt hidden states; downstream denoising passes compound the error — the policy commits to failure with no detect-or-correct mechanism.

FAPS is the first *triggered, learned*, safety-metriced activation-space shield: fires only when calibrated risk exceeds a threshold, injects a flow-model correction conditioned on failure type, falls back safely when inactive — no weights modified.

Related Work

Activation steering — RepE, ITI, ROME, MEMIT steer LLMs via additive residuals or null-space edits. None address continuous-action VLAs.

VLA robustness — VLA-OS, Don't Blind the VLA, LAE, Cerebellar, ReSteer: layer analysis, activation intervention, action-space correction, steerability. None are triggered or calibrated.

Concurrent — **COAST**: always-on closed-form conceptor steering ($h' = h \cdot M$). FAPS adds triggering, a learned flow model, and adversarial eval.

System Architecture

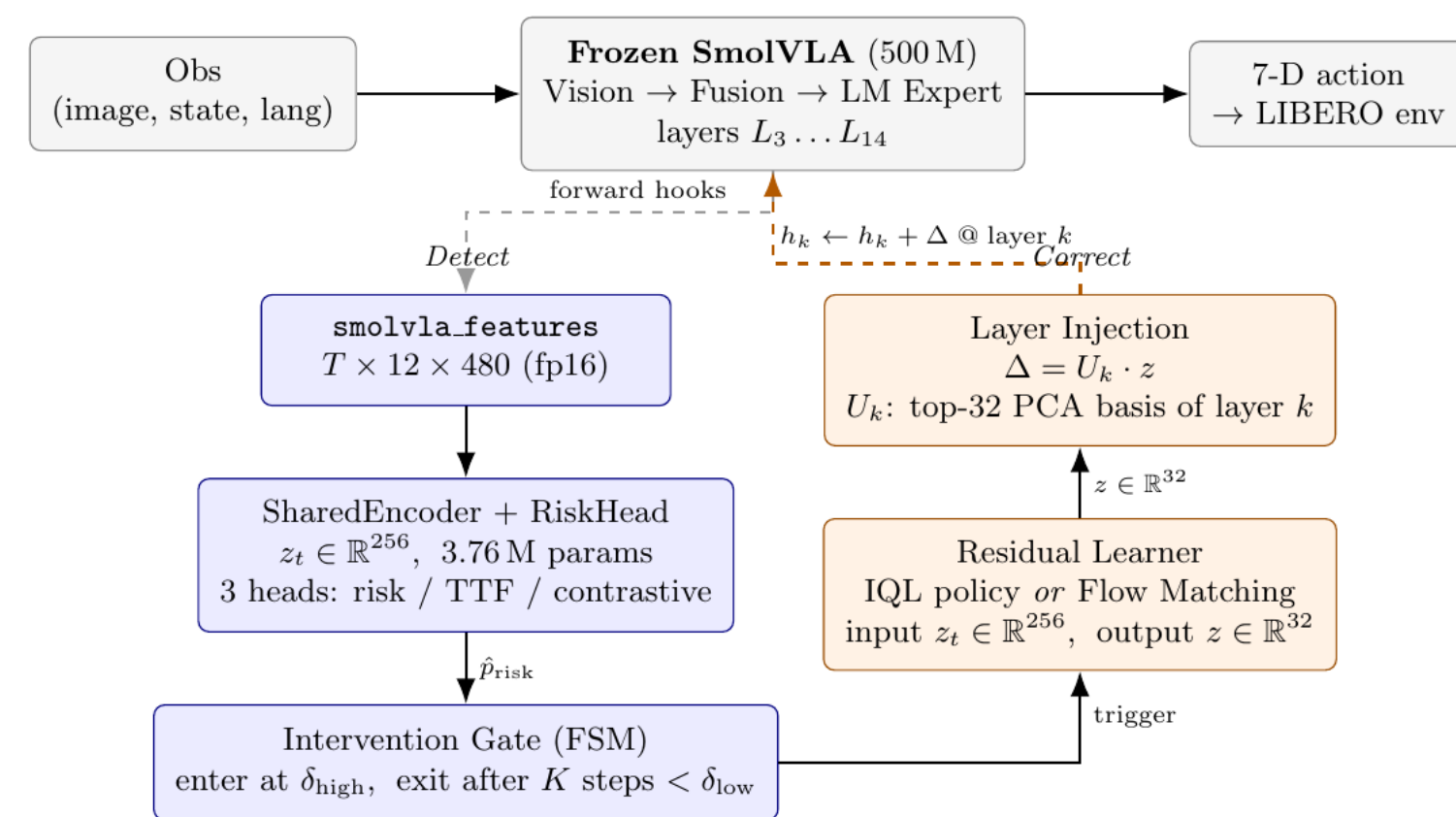


Figure 1. FAPS pipeline (inference-time). Hooks at SmolVLA layers {4, 5, 6, 11, 12, 13} feed the risk head ($\hat{p}_{\text{risk}}, z_t$); the hysteresis FSM gates injection; flow model adds $\Delta = U z$ at layer k . No VLA weights modified.

Data & Perturbations

5,117 labeled rollouts of SmolVLA on LIBERO across 11 perturbation types. Labels: fail_in_H ($H = 30$) and failure-type tag (collision, drop, grasp misalignment, grasp precision).

- **Eval only**: joint-angle noise (hard negatives)
- **Train+eval**: action noise, viewpoint shift, Gaussian blur, windowed variants ($\times 2$ each)
- **Trigger retrain**: 999 windowed perturbations + 250 nominal successes; timeout nominal failures excluded

Risk Head

SharedEncoder (3.76M): temporal CNN + cross-layer attention over 6 layers $\rightarrow \hat{p}_{\text{risk}}$ and $z_t \in \mathbb{R}^{256}$.

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{TTF}} \mathcal{L}_{\text{TTF}} + \lambda_{\text{NT-Xent}} \mathcal{L}_{\text{NT-Xent}}$$

BCE supervises risk; TTF adds time-to-failure granularity; NT-Xent induces failure-type geometry in z_t (silhouette +0.168).

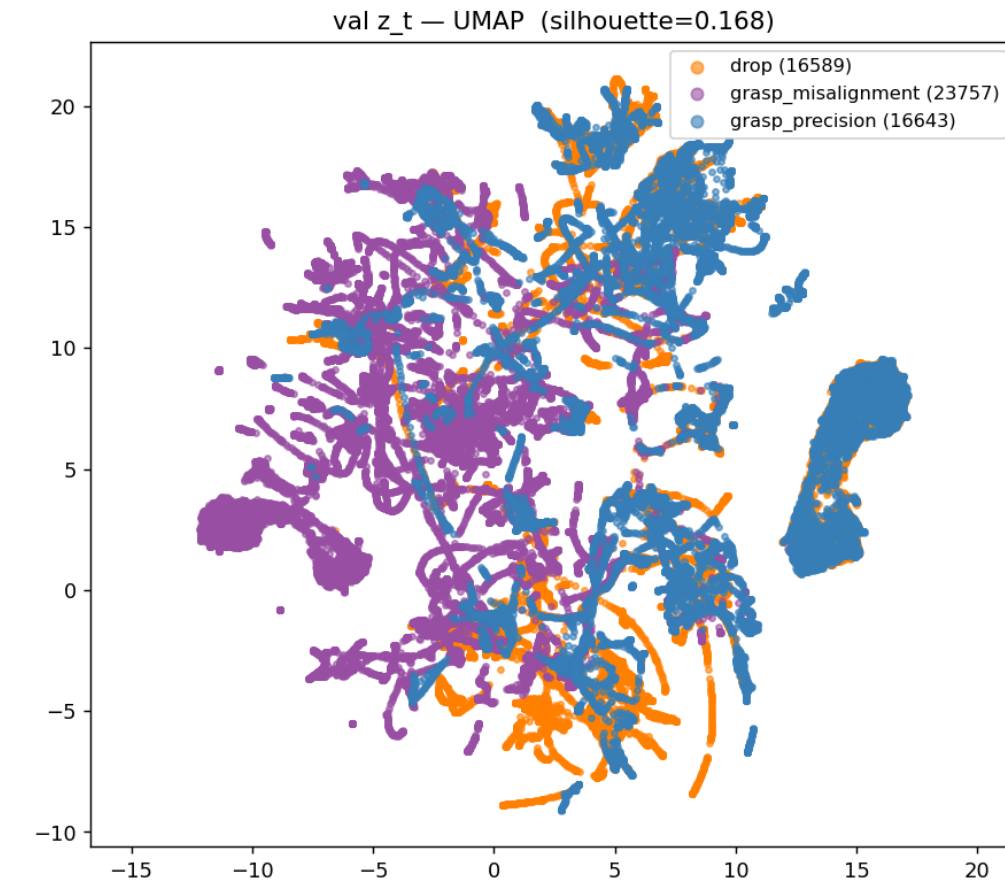


Figure 2. UMAP of z_t (Run C). Failure types cluster clearly (silhouette +0.168).

Flow-Matching Residual & FSM

Flow model conditioned on z_t generates $z \in \mathbb{R}^{48}$; projected via top-48 PCA basis U_k ($\sim 95\%$ variance): $\Delta = U_k z$, $h'_k = h_k + \Delta$.

Layer candidates: C12/C13 conceptor+failure flow runs produce substantial residuals (loss ~ 0.30 , $\|\Delta\| \sim 1.5$, 93% non-zero). Layer-11 closed-loop eval below isolates the trigger failure mode before retrain.

Hysteresis FSM: enter at $\hat{p} \geq 0.7$; exit after $K = 5$ steps below 0.4; any fault $\rightarrow$ unshielded fallback.

Risk Head Ablation

Table 1. Loss ablation. BCE detects risk; NT-Xent adds geometry.

Run	wAUC	Silh.	ECE	BSkill
A (BCE only)	0.928	-0.046	0.088	-0.067
B (BCE+TTF)	0.927	-0.040	0.093	-0.12
C (BCE+TTF+NT-Xent)	0.917	+0.168	0.077	+0.018
D (NT-Xent only)	0.440	+0.125	—	—

Run C selected. NT-Xent costs -0.010 wAUC vs. B, raises silhouette by $+0.208$, improves calibration (ECE $0.093 \rightarrow 0.077$).

Layer Probe

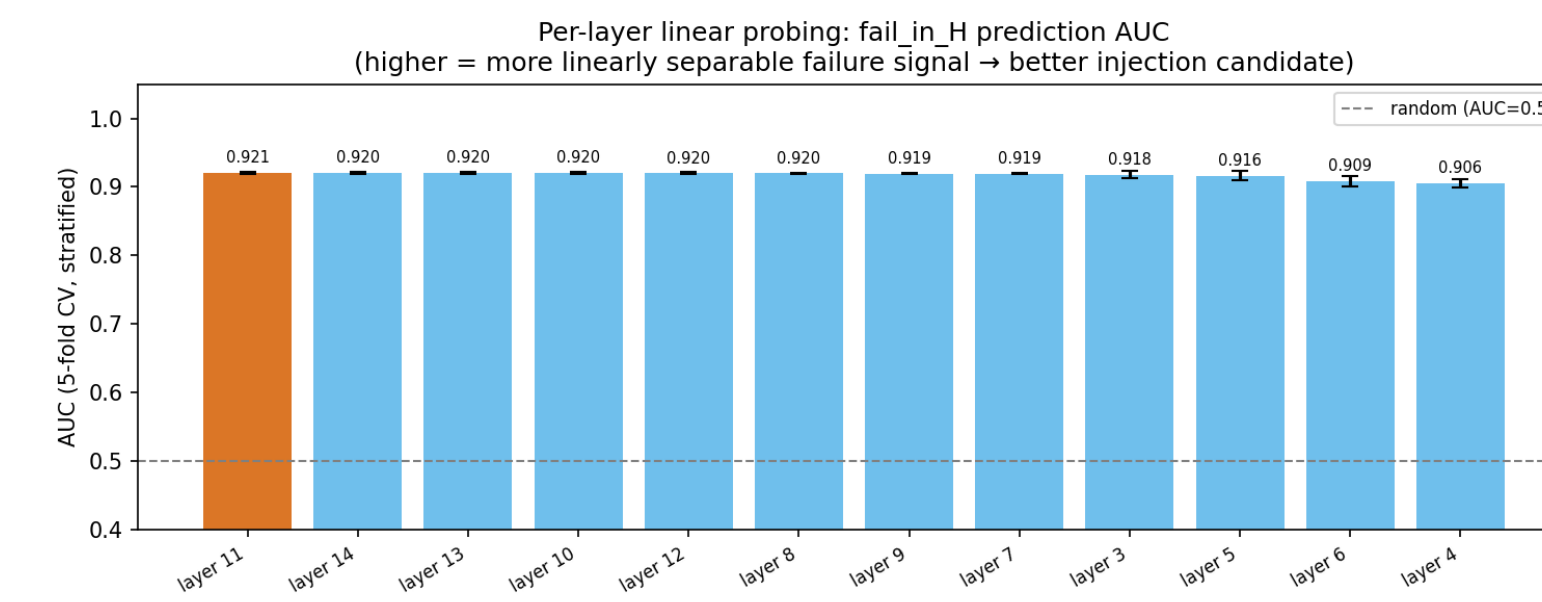


Figure 3. Linear probe AUC per action-expert layer (≈ 0.92 flat across all 12). Failure is linearly separable everywhere in SmolVLA's activations. Layer 11 (AUC 0.921) selected: mid-depth lets downstream layers regenerate coherent actions from the corrected state. Final layer comparison via closed-loop SR delta is in progress.

Shield Evaluation

Pre-retrain eval: 80 closed-loop LIBERO episodes using the Run C risk head and layer-11 injection. This run exposed the bottleneck: the residual correction works when the trigger is precise, but false positives cause over-injection on clean steps.

Table 2. Pre-retrain shield eval. Positive SR delta is possible, but false-positive injection precision is still low.

Perturbation	Unsh. SR	FAPS SR	Δ SR	Prec.	FPR
none	0.60	0.60	0.00	—	—
action noise	0.70	0.80	+0.10	0.183	0.896
image blur	0.80	0.80	0.00	0.165	0.936
joint noise	0.80	0.70	-0.10	0.104	0.782

Diagnostic split: perturbed shielded episodes with no false positives succeeded 17/17; episodes with at least one false-positive injection succeeded only 46%. Phase 1 retrain adds nominal successes as hard negatives to reduce this false-trigger rate.

Discussion & Takeaways

Risk is accessible in VLA activations. BCE+TTF reaches wAUC 0.927; adding contrastive supervision preserves detector quality while making the representation useful for correction.

Contrastive geometry is the key design choice. NT-Xent raises silhouette $-0.040 \rightarrow +0.168$ at only -0.010 wAUC cost — enabling the flow model to condition on failure type.

The current online bottleneck is calibration. When the trigger fires only inside the perturbation window, shielded episodes succeed; failures concentrate in false-positive over-injection.

Limitations

- Simulation only (LIBERO); real-world not validated.
- Single VLA (SmolVLA 500M).
- Poster numbers currently use the pre-retrain risk head.
- Final C12/C13 shield eval depends on retrained trigger calibration.
- PCA basis U_k vs. null-space basis U_{null} not yet compared.

Future Directions

- **U_{null} injection**: null-space projection guarantees zero nominal degradation by construction.
- **Multi-task**: one risk head and flow model across task suites and VLA families.
- **Real-world transfer**: physical hardware with realistic sensor noise.
- **Online adaptation**: test-time risk head update via lightweight meta-learning.

Full References & Appendix



Scan for full references, additional figures, and experiment appendix.

github.com/equalsdaniel/CS224R_robotics_project